Advaith Kandiraju

1)

$$ {}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

α_{i-1}	a_{i-1}	d_i	θ_i
a_1	0	0	θ_1
a_2	0	0	θ_2
a_3	0	0	θ_3

$$ {}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & a_2 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & a_3 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^0_3T = \begin{array}{cl} \begin{bmatrix} c_{123} & -s_{123} & 0 & a_1 + a_2 c_1 + a_3 c_{12} \\ s_{123} & c_{123} & 0 & a_2 s_1 + a_3 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \begin{array}{l} \to x \\ \to y \\ \to z \\ \\ \end{array} \end{array} $$

$$ {}^0_nJ_{lin} = \begin{bmatrix} \partial x/\partial\theta_1 & \partial x/\partial\theta_2 & \partial x/\partial\theta_3 \\ \partial y/\partial\theta_1 & \partial y/\partial\theta_2 & \partial y/\partial\theta_3 \\ \partial z/\partial\theta_1 & \partial z/\partial\theta_2 & \partial z/\partial\theta_3 \end{bmatrix} $$

$$ {}^0_nJ_{avg} = \left[{}^0_1R\,{}^1Z_1 \quad {}^0_2R\,{}^2Z_2 \quad {}^0_3R\,{}^3Z_3 \right] $$

$$ {}^0_nJ = \begin{bmatrix} -a_2 s_1 - a_3 s_2 & -a_3 s_{12} & 0 \\ a_2 c_1 + a_3 c_{12} & a_3 c_{12} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} $$

2)

$${}^{0}_{1}T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

a_{i-1}	α_{i-1}	d_i	θ_i
0	0	0	θ_1
a_2	0	0	θ_2
a_3	0	0	θ_3

$${}^{1}_{2}T = \begin{bmatrix} c_3 & -s_3 & 0 & a_3 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{0}_{3}T = \begin{bmatrix} c_{123} & -s_{123} & 0 & a_2 c_1 + a_3 c_{12} \\ 0 & 0 & -1 & 0 \\ s_{123} & c_{123} & 0 & a_2 s_1 + a_3 s_{12} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{0}_{n}J_{lin} = \begin{bmatrix} \partial x/\partial\theta_1 & \partial x/\partial\theta_2 & \partial x/\partial\theta_3 \\ \partial y/\partial\theta_1 & \partial y/\partial\theta_2 & \partial y/\partial\theta_3 \\ \partial z/\partial\theta_1 & \partial z/\partial\theta_2 & \partial z/\partial\theta_3 \end{bmatrix}$$

$${}^{0}_{n}J_{avg} = \begin{bmatrix} {}^{0}_{1}R\,{}^{1}Z_1 & {}^{0}_{2}R\,{}^{2}Z_2 & {}^{0}_{3}R\,{}^{3}Z_3 \end{bmatrix}$$

$${}^{0}_{n}J = \begin{bmatrix} -a_2 s_1 - a_3 s_{12} & a_3 s_{12} & 0 \\ 0 & 0 & 0 \\ a_2 c_1 + a_3 c_{12} & a_3 c_{12} & 0 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

3)

$${}^1w_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad v_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$${}^2w_2 = {}^2_1R\,{}^1w_1 + \begin{bmatrix} 0 \\ 0 \\ \theta_2 \end{bmatrix}$$

$${}^2w_2 = \begin{bmatrix} c_2 & 0 & s_2 \\ -s_2 & 0 & c_2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \theta_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \theta_2 \end{bmatrix}$$

$${}^2w_2 = \begin{bmatrix} s_2\dot{\theta}_1 \\ c_2\dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \qquad {}^2v_2 = \left({}^2_1R\ {}^1v_1 + {}^1w_1 \times {}^1P_2\right)$$

$${}^2v_2 = \begin{bmatrix} c_2 & 0 & s_2 \\ -s_2 & 0 & c_2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ L_1\theta_1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -L_1\theta_1 \end{bmatrix}$$

$$ {}^{3}w_3 = {}^{3}_{2}R\,{}^{2}w_2 + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix} $$

$$ {}^{3}w_3 = \begin{bmatrix} s_{23}\dot{\theta}_1 \\ c_{23}\dot{\theta}_1 \\ \dot{\theta}_2 + \dot{\theta}_3 \end{bmatrix} \qquad {}^{3}v_3 = \left({}^{3}_{2}R \quad {}^{2}v_2 + {}^{2}w_2 \times {}^{2}P_3 \right) $$

$$ {}^{3}v_3 = \begin{bmatrix} c_3 & s_3 & 0 \\ -s_3 & c_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \left[\begin{pmatrix} 0 \\ 0 \\ -L_1\dot{\theta}_1 \end{pmatrix} + \begin{pmatrix} 0 \\ L_2\dot{\theta}_2 \\ -L_2 c_2 \dot{\theta}_1 \end{pmatrix} \right] $$

$$ {}^{3}v_3 = \begin{bmatrix} s_3 L_2 \dot{\theta}_2 \\ c_3 L_2 \dot{\theta}_2 \\ -L_1\dot{\theta}_1 - L_2 c_2 \dot{\theta}_1 \end{bmatrix} $$

$$ {}^{4}w_4 = {}^{3}w_3 $$

$$ {}^{4}v_4 = {}^{4}_{3}R \left({}^{3}v_3 + {}^{3}w_3 \times {}^{3}P_4 \right) $$

$$ = \begin{pmatrix} s_2 L_2 \dot{\theta}_2 \\ c_3 L_2 \dot{\theta}_2 \\ -L_1\dot{\theta}_1 - L_2 c_2 \dot{\theta}_1 \end{pmatrix} + \begin{bmatrix} 0 \\ L_3(\dot{\theta}_2 + \dot{\theta}_3) \\ -L_3 c_{23} \dot{\theta}_1 \end{bmatrix} $$

$$ = \begin{bmatrix} s_3 L_2 \dot{\theta}_2 \\ c_3 L_2 \dot{\theta}_2 - L_3(\dot{\theta}_2 + \dot{\theta}_3) \\ -L_1\dot{\theta}_1 - L_2 c_2 \dot{\theta}_1 - L_3 c_{2S}\dot{\theta}_1 \end{bmatrix} $$

$$ {}^{4}J_\theta = \begin{bmatrix} 0 & s_3 L_2 & 0 \\ 0 & c_3 L_2 + L_3 & L_3 \\ (-L_1 + L_2 c_2 + L_3 L_{23}) & 0 & 0 \end{bmatrix} $$

4) $\tau = {}^3J^{T\,3}F$

$${}^3F = {}^3J^{-1}\tau$$

$${}^1w_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \qquad {}^1v_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$${}^2w_2 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \qquad {}^2v_2 = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ l_1\dot{\theta}_1 \\ 0 \end{bmatrix}$$

$${}^2v_2 = \begin{bmatrix} l_1 s_2 \dot{\theta}_1 \\ l_2 c_2 \dot{\theta}_1 \\ 0 \end{bmatrix}$$

$${}^3w_3 = {}^2w_2$$

$${}^3v_3 = \begin{bmatrix} l_2 s_2 \dot{\theta}_1 \\ l_1 c_2 \dot{\theta}_1 + l_2 (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}$$

$${}^3J = \begin{bmatrix} l_1 s_2 & 0 \\ l_1 c_2 + l_2 & l_2 \end{bmatrix}$$

$${}^3J^T = \begin{bmatrix} l_1 s_2 & l_1 c_2 + l_2 \\ 0 & l_2 \end{bmatrix}$$

$$\boxed{{}^3J^{-1} = \frac{1}{l_1 l_2 s_2} \begin{bmatrix} l_2 & -l_1 c_1 - l_2 \\ 0 & l_1 s_2 \end{bmatrix}}$$